

A Provocative Re-Evaluation of Einsteinian Axioms

An Assumption-Stress-Testing Framework for Speculative Foundations of Physics

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Abstract

Modern physics is extraordinarily successful, yet its success does not automatically imply that every assumption underlying its most successful theories is fundamental. This paper develops a deliberately provocative but technically conservative framework for examining this distinction.

The central methodological proposal is called *Assumption Stress Testing* (AST): assumptions are separated from mathematical consequences, physical predictions, experimental evidence, and demonstrated domains of validity.

The framework is applied primarily to special relativity and general relativity, with particular attention to Lorentz invariance, the equivalence principle, spacetime geometry, general covariance, gravitation, singularities, black holes, cosmology, and the unresolved relationship between gravity and quantum theory.

Einstein's work is treated not as a collection of untouchable axioms, but as a sequence of historically successful conceptual constructions whose empirical consequences can be separated from the assumptions used to obtain them.

A speculative research direction is then introduced in which spacetime geometry may be interpreted as an emergent relational structure associated with underlying informational degrees of freedom. This proposal is not presented as an established physical theory. Instead, it is formulated as a research program subject to a stringent conservative-extension requirement: any deeper theory must reproduce established relativistic physics in every experimentally validated regime unless it explains a confirmed discrepancy.

The paper therefore distinguishes three propositions that are frequently conflated: that a theory is successful, that a theory is fundamental, and that a theory is complete.

The central claim is methodological rather than revolutionary:

Successful $\neq$ Fundamental

while

Incomplete $\neq$ Wrong.

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1 Introduction

Physics advances by constructing mathematical descriptions that connect assumptions to observations. The most successful theories are not merely collections of equations. They are frameworks containing principles, mathematical structures, approximations, interpretations, and domains of validity.

Einstein’s theories of relativity provide an unusually clear example. Special relativity transformed the relationship between space and time by making Lorentz symmetry and the invariant speed of light central components of relativistic kinematics.

General relativity subsequently transformed gravitation from a force-like interaction into a geometric description of spacetime dynamics.

Both theories have survived an extraordinary range of experimental tests. Their success is therefore not reasonably characterized as evidence that they are poor theories. Quite the opposite is true. They are among the most empirically successful theories ever developed.

Nevertheless, empirical success and ontological finality are different propositions.

A theory can correctly predict every phenomenon within a particular experimentally accessible domain while still being an effective description of a deeper structure.

Classical mechanics provides a familiar historical example. Newtonian mechanics remains highly useful in its domain even though it is not the fundamental description of relativistic or quantum systems.

The same logical possibility motivates the present work.

The title of this paper is intentionally irreverent. The mathematics, however, is intended to be treated seriously. The phrase “utter batsh*t insanity” is not meant to imply that relativity is

irrational. Instead, it describes the experience of deliberately pushing foundational assumptions to their conceptual limits.

The resulting question is:

Which parts of modern relativistic physics are empirically unavoidable, which are mathematical choices, which are interpretive structures, and which might emerge from something deeper?

This paper develops a framework for asking that question without confusing speculation with established physics.

2 Assumptions, Mathematics, Predictions, and Evidence

A common failure mode in theoretical discussions is to collapse several logically distinct objects into one.

Consider the chain

Assumptions $\longrightarrow$ Mathematical Structure $\longrightarrow$ Physical Predictions $\longrightarrow$ Experimental Evidence.

These arrows are not reversible.

Experimental evidence can strongly constrain a mathematical theory, but an experiment does not necessarily determine a unique ontology. Multiple mathematical descriptions can sometimes encode the same observable physics.

For this reason, this paper introduces the following representation of a theory:

$$\mathcal{T} = (A, M, D, P, E),$$

where

$$A = \text{foundational assumptions}, \tag{1}$$

$$M = \text{mathematical structure}, \tag{2}$$

$$D = \text{derived dynamics}, \tag{3}$$

$$P = \text{physical predictions}, \tag{4}$$

$$E = \text{empirical evidence}. \tag{5}$$

The tuple is conceptual rather than mathematical in the strict sense.

Its purpose is bookkeeping.

A successful theory should therefore be examined at several levels:

1. What does the theory assume?
2. What mathematical structure follows from those assumptions?
3. What predictions follow from that mathematical structure?
4. Which predictions have actually been tested?
5. Over what range of conditions have they been tested?

6. Which components are logically necessary and which are merely convenient?
7. Could an alternative theory reproduce the same validated predictions?

This distinction becomes particularly important when discussing the possibility of deeper theories.

3 Assumption Stress Testing

The central methodological idea of this paper is called *Assumption Stress Testing* (AST).

The process may be represented schematically as

$$A \rightarrow M \rightarrow P \rightarrow E \rightarrow D,$$

where

$$A : \text{assumptions}, \tag{6}$$

$$M : \text{mathematical consequences}, \tag{7}$$

$$P : \text{physical predictions}, \tag{8}$$

$$E : \text{experimental evidence}, \tag{9}$$

$$D : \text{demonstrated domain of validity}. \tag{10}$$

The purpose of AST is not to attack successful theories merely for entertainment. Its purpose is to identify which assumptions are empirically constrained and which remain open to reinterpretation.

For example, if a particular assumption is modified but every experimentally tested prediction remains unchanged, then the assumption may not be uniquely fixed by current observations.

Conversely, if modifying the assumption immediately produces an experimentally excluded prediction, then the assumption is strongly constrained.

This produces three broad categories:

1. **Empirically constrained assumptions:** modifications produce experimentally excluded consequences.
2. **Underdetermined assumptions:** multiple formulations produce the same currently observed consequences.
3. **Speculative assumptions:** assumptions introduced beyond currently established evidence.

The third category must not be confused with discovery.

A speculative hypothesis becomes physically significant only when it develops quantitative consequences that can be compared with observation.

4 The Assumption Budget

A useful conceptual tool is an “assumption budget.”

Let

$$\mathcal{B} = \sum_i w_i A_i,$$

where A_i represents an assumption and w_i represents a conceptual weight assigned to it. This quantity is not a physical observable.

It is a bookkeeping device intended to ask whether a proposed explanation introduces substantially more unexplained structure than the phenomenon it is supposed to explain.

The idea resembles a principle of theoretical economy, but it should not be interpreted as a rigid numerical criterion.

For example, a theory containing

$$A_1, A_2, A_3, A_4$$

may appear less economical than one containing

$$B_1, B_2,$$

but the comparison is meaningful only if the theories have comparable explanatory power.

A simpler theory that cannot reproduce observations is not preferable merely because it has fewer assumptions.

Thus,

$$\text{simplicity} \neq \text{correctness}.$$

Likewise,

$$\text{complexity} \neq \text{incorrectness}.$$

The objective is explanatory compression without sacrificing empirical adequacy.

5 Einstein's Special Relativity

Einstein's 1905 paper "On the Electrodynamics of Moving Bodies" established the modern formulation of special relativity [1].

Two central principles are commonly emphasized:

1. The laws of physics take the same form in inertial reference frames.
2. The speed of light in vacuum is invariant for inertial observers.

These principles lead to the Lorentz transformation.

For two inertial frames related by a relative velocity v along the x -direction,

$$x' = \gamma(x - vt), \tag{11}$$

$$t' = \gamma \left(t - \frac{vx}{c^2} \right), \tag{12}$$

where

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

The transformation preserves the spacetime interval

$$s^2 = -c^2t^2 + x^2 + y^2 + z^2.$$

In differential form,

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2.$$

The invariance of this quantity is more fundamental than the familiar statements concerning time dilation and length contraction.

Time dilation and length contraction are consequences.

They are not independent axioms.

This illustrates the importance of AST: the phenomenological statements frequently taught first are not necessarily the deepest mathematical structure.

6 Mass-Energy Equivalence

Einstein's 1905 discussion of the dependence of inertia on energy content led to the famous relation

$$E_0 = mc^2,$$

where E_0 denotes rest energy [2].

The more general relativistic energy-momentum relation is

$$E^2 = p^2c^2 + m^2c^4.$$

For a particle at rest,

$$p = 0,$$

and therefore

$$E = mc^2.$$

The importance of this relation extends beyond the numerical conversion between mass and energy.

It indicates that quantities previously treated as conceptually distinct can belong to a deeper relativistic structure.

This raises a broader methodological question:

If mass can be understood as one manifestation of a deeper physical quantity, could other apparently fundamental structures also be emergent?

This question does not imply that mass is merely “information.” It simply motivates investigation into whether apparently primitive physical quantities have deeper descriptions.

7 From Special to General Relativity

Einstein's work on gravitation gradually developed from the special-relativistic principle that the laws of physics should not privilege inertial observers.

The equivalence principle became a central conceptual ingredient.

At its simplest, the weak equivalence principle concerns the universality of free fall.

A hierarchy can be introduced:

Weak equivalence principle $\longrightarrow$ Einstein equivalence principle $\longrightarrow$ geometric formulation.

The Einstein equivalence principle incorporates local Lorentz invariance and local position invariance together with the universality of free fall.

General relativity takes the next conceptual step by representing gravitation through spacetime geometry.

8 General Relativity

Einstein's field equations can be written as

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

Here,

$$G_{\mu\nu} : \text{Einstein tensor}, \tag{13}$$

$$g_{\mu\nu} : \text{metric tensor}, \tag{14}$$

$$\Lambda : \text{cosmological constant}, \tag{15}$$

$$G : \text{Newton's gravitational constant}, \tag{16}$$

$$c : \text{speed of light}, \tag{17}$$

$$T_{\mu\nu} : \text{stress-energy tensor}. \tag{18}$$

The geometric side is constrained by the Bianchi identity,

$$\nabla^\mu G_{\mu\nu} = 0.$$

Therefore, for constant Λ ,

$$\nabla^\mu T_{\mu\nu} = 0.$$

The equation expresses a profound relationship:

$$\text{spacetime geometry} \longleftrightarrow \text{matter and energy}.$$

The structure of general relativity therefore combines:

1. differential geometry,
2. local Lorentz symmetry,

3. causal structure,
4. dynamical gravitational fields,
5. conservation laws,
6. and matter coupling.

Any proposed deeper theory must recover these structures in the appropriate classical limit.

9 General Covariance

General covariance is frequently misunderstood.

A coordinate transformation can change the numerical values of tensor components without changing the underlying geometric object.

If

$$x^\mu \rightarrow x'^\mu(x),$$

then tensor components transform according to the appropriate tensor transformation laws.

The physical content of the theory therefore cannot simply be identified with a particular coordinate representation.

This motivates a distinction:

$$\text{coordinate description} \neq \text{physical ontology}.$$

The distinction becomes especially important when considering emergent spacetime.

If spacetime is emergent, then coordinates may represent relationships within an effective description rather than fundamental labels existing independently of physical systems.

10 Operational Spacetime

Consider a hypothetical hierarchy

$$\mathcal{I} \rightarrow \mathcal{R} \rightarrow g_{\mu\nu}.$$

Here,

$$\mathcal{I} : \text{microscopic informational degrees of freedom}, \quad (19)$$

$$\mathcal{R} : \text{relations between physical degrees of freedom}, \quad (20)$$

$$g_{\mu\nu} : \text{effective spacetime geometry}. \quad (21)$$

This is not proposed as an established result.

It is a conceptual research hypothesis.

The key idea is that geometry may describe relationships between physical systems rather than existing as an entirely independent primitive object.

Operationally, a metric is inferred through measurements involving clocks, rods, signals, particles, and fields.

One may therefore define a set of operational observables

$$\mathcal{O} = \{O_1, O_2, \dots, O_N\}$$

and relational quantities

$$\mathcal{R}_{ij} = f(O_i, O_j).$$

If a sufficiently smooth macroscopic structure emerges, one could imagine a mapping

$$\mathcal{R} \longrightarrow g_{\mu\nu}.$$

The difficult problem is then not writing the metric.

The difficult problem is deriving the metric.

11 Metric as an Information Structure

Suppose microscopic states are represented by variables

$$\chi = \{\chi_1, \chi_2, \dots, \chi_N\}.$$

A microscopic theory might possess an action

$$S_{\text{micro}}[\chi].$$

A coarse-graining operation could produce effective variables

$$\chi \rightarrow \chi_{\text{eff}}.$$

If spacetime geometry is emergent, one might seek a functional

$$g_{\mu\nu} = \mathcal{C}_{\mu\nu}[\chi],$$

where $\mathcal{C}_{\mu\nu}$ represents a coarse-graining or reconstruction map.

The resulting effective action might then have the schematic form

$$S_{\text{eff}}[g] = \mathcal{C} [S_{\text{micro}}[\chi]].$$

The challenge is enormous.

The map must reproduce:

1. Lorentzian signature,
2. causal structure,
3. approximate locality,
4. diffeomorphism invariance,
5. Einsteinian dynamics,
6. quantum-to-classical behavior,
7. and the observed coupling to matter.

A proposal that reproduces only one of these features is insufficient.

12 Geodesic Motion

In general relativity, freely falling massive particles follow timelike geodesics:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0.$$

The Christoffel symbols are

$$\Gamma_{\alpha\beta}^\mu = \frac{1}{2} g^{\mu\nu} \left(\frac{\partial g_{\nu\alpha}}{\partial x^\beta} + \frac{\partial g_{\nu\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\nu} \right).$$

The geodesic equation illustrates a striking feature of general relativity.

Gravity does not need to appear as a conventional force term.

Instead, the geometry determines inertial motion.

This creates a conceptual possibility:

$$\text{gravity} \approx \text{geometry of inertial motion}.$$

A deeper theory would have to explain why this geometric description emerges so accurately.

13 Geodesic Deviation and Curvature

The physical distinction between a gravitational field and a locally inertial coordinate system is captured by curvature.

For a family of nearby geodesics, the separation vector ξ^μ satisfies

$$\frac{D^2 \xi^\mu}{D\tau^2} = -R^\mu_{\nu\alpha\beta} u^\nu \xi^\alpha u^\beta.$$

Here,

$$R^\mu_{\nu\alpha\beta} : \text{Riemann curvature tensor}, \quad (22)$$

$$u^\mu : \text{four-velocity}, \quad (23)$$

$$\xi^\mu : \text{separation vector}. \quad (24)$$

This suggests an important distinction:

$$\text{coordinate acceleration} \neq \text{spacetime curvature}.$$

A coordinate system can introduce apparent acceleration.

Curvature cannot generally be removed throughout an extended region by a coordinate transformation.

14 Causal Structure

The metric does more than define distances.

It determines causal relationships.

For a displacement dx^μ ,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

Using the sign convention adopted here,

$$ds^2 < 0$$

is timelike,

$$ds^2 = 0$$

is null, and

$$ds^2 > 0$$

is spacelike.

This raises another foundational question:

Could causal structure be more fundamental than metric structure?

A hypothetical theory might begin with a causal relation

$$x \prec y$$

meaning that event x can causally precede event y .

A sufficiently rich causal structure may contain substantial information about the underlying geometry.

The speculative hierarchy would therefore become

$$\text{microscopic relations} \rightarrow \text{causal structure} \rightarrow \text{metric geometry}.$$

Again, this is a research hypothesis rather than an established conclusion.

15 Operational Relationality

The preceding ideas motivate a possible principle:

Physical geometry should ultimately be expressible in terms of measurable relations among physical systems.

Call this the *Operational Relationality Principle*.

The principle does not assert that reality is “only information.”

Rather, it asks whether the quantities used to describe spacetime can be reconstructed from relationships among physical degrees of freedom.

Information without a physical carrier is an abstraction.

Therefore, any information-theoretic formulation must specify:

1. what stores the information,
2. what transformations act upon it,
3. what observables can be measured,
4. how locality is defined,
5. and how classical spacetime emerges.

Without these ingredients, “information” risks becoming a replacement word for “something we do not yet understand.”

16 The Einstein-Hilbert Action

The dynamics of general relativity can be derived from the Einstein-Hilbert action

$$S_{\text{EH}} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda).$$

Here,

$$g = \det(g_{\mu\nu})$$

and R is the Ricci scalar.

Varying the action with respect to the metric yields Einstein's field equations.

This action-based formulation is especially useful for discussing possible extensions.

A generalized effective action could take the form

$$S_{\text{eff}} = S_{\text{EH}} + S_{\text{matter}} + S_{\text{corrections}}.$$

For example,

$$S_{\text{corrections}} = \int d^4x \sqrt{-g} \left[\alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} + \dots \right].$$

The coefficients

$$\alpha, \beta, \gamma$$

would encode deviations from the simplest Einstein-Hilbert description.

Any such corrections must be sufficiently suppressed in regimes where general relativity has already been experimentally verified.

17 The Conservative-Extension Principle

A candidate fundamental theory should satisfy

$\text{New theory} \longrightarrow \text{Established theory}$

in the appropriate limit.

More explicitly,

$$\mathcal{T}_{\text{new}} \xrightarrow[\text{validated regime}]{} \mathcal{T}_{\text{established}}.$$

For a proposed successor to general relativity,

$$\mathcal{T}_{\text{new}} \rightarrow \text{GR}$$

should occur wherever general relativity has been successfully tested.

Likewise, a deeper theory of relativistic physics must reproduce special relativity when gravitational curvature is negligible.

This prevents a common mistake in speculative physics:

$$\text{interesting alternative} \neq \text{successful replacement}.$$

A theory that contradicts established experiments without explaining those contradictions is not an improvement merely because its conceptual structure is more exotic.

18 Einstein and the Search for Deeper Unification

Einstein's own career demonstrates that scientific theories can be extraordinarily successful while their underlying conceptual structures remain open to extension.

After developing general relativity, Einstein continued investigating unified descriptions of physics.

His later efforts did not produce an experimentally established unified field theory, but they demonstrate an important methodological point:

questioning foundations $\neq$ rejecting established evidence.

Einstein questioned the structure of physics while preserving the empirical successes that motivated the questions.

This provides a useful model for modern foundational research:

preserve what works + investigate what remains unexplained.

19 Singularities as Diagnostic Objects

General relativity contains solutions in which curvature invariants can become unbounded or geodesic completeness can fail.

Such situations are commonly described using the language of “singularities.”

However, a singularity should not automatically be interpreted as a literal physical object with infinite density.

A more conservative interpretation is

singularity $\rightarrow$ possible breakdown of the effective description.

This does not establish that general relativity is wrong.

It establishes that classical general relativity may not provide a complete description of every conceivable regime.

Thus,

Incomplete $\neq$ Wrong

20 Black Holes and Information

Black holes represent an important intersection between gravitation, thermodynamics, and quantum theory.

The Bekenstein-Hawking entropy is proportional to horizon area:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar}.$$

The appearance of the constants

$$k_B, \quad c, \quad G, \quad \hbar$$

in one equation combines thermodynamics, quantum theory, and gravity.

This motivates questions about whether spacetime geometry, information, and quantum degrees of freedom are connected at a deeper level.

However, the existence of such connections does not by itself prove that spacetime is informational. They provide motivation for further investigation.

21 Cosmological Constant and Vacuum Structure

The cosmological constant appears in Einstein's equations as

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

It contributes to the large-scale dynamics of spacetime.

The relationship between vacuum energy, quantum field theory, and cosmological observations raises deep questions concerning the relationship between quantum theory and gravitational dynamics.

The present paper does not attempt to solve this problem.

Instead, it identifies it as an example of a broader issue:

successful effective description may coexist with unresolved microscopic interpretation.

22 Is Lorentz Invariance Fundamental?

Lorentz invariance is one of the most thoroughly tested structural features of modern physics.

This does not logically prove that Lorentz invariance must be fundamental at every conceivable scale.

It is possible, in principle, that a deeper theory possesses microscopic variables whose collective behavior produces approximate Lorentz symmetry.

The hierarchy would be

microscopic dynamics $\rightarrow$ emergent Lorentz symmetry $\rightarrow$ observed relativistic physics.

This possibility must not be confused with experimentally established Lorentz violation.

An emergent symmetry can be exact to extraordinary precision over accessible energy scales.

The important question is therefore:

Does Lorentz symmetry have to be fundamental, or can it emerge?

A meaningful answer would require a concrete model.

23 A Possible Information-Geometric Emergence Program

The ideas developed above can be combined into a speculative research program:

$$\boxed{\mathcal{I}_{\text{fundamental}} \rightarrow \mathcal{R}_{\text{relational}} \rightarrow g_{\mu\nu} \rightarrow \text{classical spacetime}}$$

The proposed stages are:

1. Fundamental degrees of freedom exist at a microscopic level.
2. Those degrees of freedom possess relational structure.
3. Coarse-graining produces an effective causal network.
4. The causal network admits an approximate metric description.
5. The emergent metric obeys an effective gravitational action.
6. Einstein's equations appear in the appropriate low-energy limit.

The key mathematical problem is therefore to construct a map

$$\mathcal{C} : \chi \mapsto g_{\mu\nu}.$$

A stronger version requires

$$\mathcal{C} [S_{\text{micro}}[\chi]] = S_{\text{EH}}[g] + \Delta S,$$

where

$$\Delta S \rightarrow 0$$

in the classical low-energy limit.

24 Requirements for an Emergent-Spacetime Theory

A credible theory of emergent spacetime would need to explain at least the following:

1. Lorentzian signature;
2. causal structure;
3. approximate locality;
4. diffeomorphism invariance;
5. Einsteinian dynamics;
6. coupling to matter;
7. quantum consistency;
8. and the classical limit.

A theory reproducing only one or two of these properties would not yet constitute a complete theory of emergent spacetime.

25 A Taxonomy of Possible Extensions

Speculative modifications of established physics can be divided into several categories.

Table 1: Conceptual taxonomy of possible extensions to relativistic physics.

Category	Central Question
Kinematic	Is Lorentz symmetry fundamental or emergent?
Geometric	Is the metric fundamental or emergent?
Dynamical	Is the Einstein-Hilbert action fundamental?
Matter coupling	Is the standard coupling between matter and geometry complete?
Quantum	What microscopic quantum degrees of freedom generate classical spacetime?
Cosmological	How do microscopic laws produce large-scale cosmic dynamics?
Informational	Can geometry be reconstructed from physical information and relations?

26 A More General Theory Space

Let a theory be represented abstractly as

$$\mathcal{T} = (A, M, D, P, E).$$

A new theory

$$\mathcal{T}' = (A', M', D', P', E')$$

may differ from the original theory at several levels.

Two formulations may use different mathematics while producing identical observable predictions:

$$P' = P, \quad M' \neq M.$$

A deeper theory may also reproduce all currently observed classical behavior while modifying physics at inaccessible scales:

$$P'_{\text{classical}} \approx P_{\text{classical}},$$

while

$$P'_{\text{microscopic}} \neq P_{\text{microscopic}}.$$

A third possibility is that a theory agrees with existing experiments while predicting deviations in extreme regimes.

This is one of the most scientifically useful possibilities because it creates a route toward empirical discrimination.

27 What Would Actually Falsify Relativity?

The phrase “falsify relativity” should be used carefully.

A disagreement between an experiment and a theoretical prediction would have to survive examination of experimental uncertainties, systematic effects, calibration, theoretical approximations, and alternative explanations.

Schematically,

$$P_{\text{experiment}} \neq P_{\text{GR}}$$

would motivate further investigation.

A stronger criterion would be

$$|P_{\text{experiment}} - P_{\text{GR}}| > \Delta_{\text{total}},$$

where Δ_{total} represents the combined relevant uncertainty.

Potential evidence against a theory could include:

1. reproducible violations of a well-tested relativistic prediction;
2. experimentally established Lorentz violation;
3. a demonstrable internal inconsistency within the theory's domain of applicability;
4. or a more comprehensive theory that reproduces successful predictions while explaining a confirmed anomaly.

By contrast, mathematical difficulty, unresolved singularities, difficulty quantizing gravity, philosophical discomfort, or the possibility of emergent spacetime do not by themselves falsify relativity.

28 Failure Versus Incompleteness

The distinction can be summarized as

Failure $\neq$ Incomplete description

A theory fails within a domain when it makes incorrect predictions there.

A theory is incomplete when it does not provide a description of a domain beyond the range in which it has been established.

Classical mechanics is not wrong because it does not contain quantum mechanics.

It is an approximation.

Likewise, general relativity may be an effective classical description even if a deeper quantum theory of gravity exists.

29 The Problem of Fundamentalness

The term “fundamental” is frequently used without a precise definition.

A useful conceptual hierarchy is

$$\text{empirical law} \subseteq \text{effective theory} \subseteq \text{fundamental theory}.$$

This notation is schematic rather than strict mathematical set inclusion.

It represents explanatory depth.

An effective theory may accurately describe a phenomenon without specifying the microscopic degrees of freedom responsible for that phenomenon.

The question for gravity is therefore:

Could spacetime geometry play a role analogous to thermodynamic variables?

If so, the metric might be an effective macroscopic variable.

This is an intriguing possibility, but it requires considerably more than analogy.

30 Coarse-Graining and Geometry

Suppose microscopic states are represented by

$$\chi_1, \chi_2, \dots, \chi_N.$$

A coarse-graining map could be written as

$$\mathcal{C} : \{\chi_i\} \rightarrow \{\phi_a\}.$$

The variables ϕ_a might represent collective degrees of freedom.

If the collective variables possess a notion of distance or causal separation, one might define an emergent metric through

$$g_{\mu\nu} = F_{\mu\nu}(\phi, \partial\phi, \dots).$$

The challenge is to derive a smooth four-dimensional manifold.

The emergence would therefore need to explain why the effective number of macroscopic spacetime dimensions is approximately

$$D_{\text{eff}} \approx 4.$$

A deeper theory could in principle possess a microscopic structure with a different effective dimensionality.

31 Emergent Locality

Locality is another major challenge.

In ordinary field theory,

$$\phi(x)$$

is associated with a spacetime point x .

But if spacetime itself emerges, the microscopic theory cannot simply assume ordinary locality from the beginning without potentially making the emergence circular.

One possible strategy would be to define locality through interaction strength.

For microscopic degrees of freedom i and j , define

$$J_{ij}$$

as an interaction weight.

An effective distance might then be related to interaction strength through a function

$$d_{ij} = f(J_{ij}).$$

The precise function would depend on the microscopic theory.

The conceptual idea is that geometric proximity could emerge from dynamical connectivity.

32 Information and Entropy

If information is incorporated into a foundational model, entropy provides a natural mathematical language.

For probabilities p_i , the Shannon entropy is

$$S = -k \sum_i p_i \ln p_i.$$

Quantum systems instead use the von Neumann entropy

$$S = -k_B \operatorname{Tr}(\rho \ln \rho),$$

where ρ is the density operator.

The appearance of entropy in gravitational physics is especially suggestive because black-hole entropy scales with area rather than volume:

$$S_{\text{BH}} \propto A.$$

This motivates questions about whether the number of physically accessible degrees of freedom in a gravitational system is fundamentally constrained by geometry.

However, entropy-area relationships do not by themselves prove that spacetime is informational. They provide motivation for further investigation.

33 A Hypothetical Emergence Criterion

Suppose a microscopic theory produces an effective metric satisfying

$$g_{\mu\nu} = \mathcal{C}_{\mu\nu}[\chi].$$

A successful emergence mechanism would need to demonstrate that

$$S_{\text{micro}}[\chi] \rightarrow S_{\text{EH}}[g]$$

under an appropriate coarse-graining transformation.

More generally,

$$S_{\text{micro}} \rightarrow S_{\text{eff}} = S_{\text{EH}} + \Delta S.$$

The classical limit requires

$$\Delta S \rightarrow 0$$

in the experimentally validated regime.

This provides a concrete research target.

Instead of merely saying

“spacetime comes from information,”

one would have to calculate the information dynamics and derive the gravitational action. That is the difference between a philosophical slogan and a physical research program.

34 Theoretical Failure Modes

A proposed foundational theory can fail in several distinct ways.

34.1 Failure of Recovery

The theory cannot reproduce special relativity or general relativity where they are experimentally validated.

34.2 Failure of Dynamics

The theory produces a metric but cannot derive physically correct dynamics.

34.3 Failure of Causality

The microscopic model permits unacceptable causal behavior.

34.4 Failure of Locality

The model cannot reproduce effective locality.

34.5 Failure of the Classical Limit

The theory cannot explain why macroscopic observers experience smooth spacetime.

34.6 Failure of Empirical Distinguishability

The theory makes no experimentally testable predictions that distinguish it from existing descriptions.

35 Prediction as the Final Gatekeeper

The ultimate requirement of a physical theory is empirical contact.

Suppose a new model introduces a parameter λ .

The model predicts

$$O(\lambda)$$

for an observable O .

If the standard theory predicts

$$O_0,$$

then the new model becomes experimentally meaningful if

$$O(\lambda) - O_0$$

is nonzero and measurable.

A successful prediction would therefore require

$$\boxed{\Delta O = O_{\text{new}} - O_{\text{standard}} \neq 0}$$

with the difference sufficiently large to be experimentally distinguishable.

36 A Research Program

The speculative framework developed here can be converted into a concrete research program.

36.1 Stage I: Recover Special Relativity

Demonstrate that the microscopic theory possesses an emergent symmetry equivalent to Lorentz invariance.

36.2 Stage II: Recover General Relativity

Derive an effective metric satisfying Einstein's equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

36.3 Stage III: Recover Quantum Matter

Show that ordinary quantum fields or an equivalent microscopic description emerge consistently.

36.4 Stage IV: Recover the Classical Limit

Demonstrate that macroscopic systems exhibit smooth spacetime geometry.

36.5 Stage V: Identify Deviations

Derive corrections

$$\Delta S$$

or modified observables.

36.6 Stage VI: Produce a Test

Identify an experiment or observation capable of distinguishing the new theory from established physics.

Only after these stages could the proposal reasonably be described as a candidate physical theory rather than a conceptual framework.

37 Discussion

Einstein's greatest conceptual contributions were not merely equations.

Special relativity changed the interpretation of time and simultaneity.

General relativity changed the interpretation of gravity.

The possibility explored here is that a future theory could change the interpretation of spacetime itself.

Such a theory might establish that the metric is not fundamental.

Or it might establish that the metric is fundamental after all.

Both outcomes would be scientifically meaningful.

The purpose of foundational research is not to force nature to agree with a preferred philosophical conclusion.

It is to determine which structures nature actually requires.

38 Limitations

This work has several important limitations.

First, the information-geometric framework introduced here is speculative.

No microscopic Hamiltonian or action has been constructed from which the proposed emergence mechanism is derived.

Second, no new experimentally verified prediction is presented.

Third, the mapping

$$\mathcal{I} \rightarrow \mathcal{R} \rightarrow g_{\mu\nu}$$

is conceptual rather than mathematically complete.

Fourth, the relationship between information, entropy, and geometry remains an active area of theoretical research and cannot be settled through analogy alone.

Fifth, the paper does not propose replacing general relativity.

General relativity remains the established classical theory of gravitation within its validated domain.

The purpose is instead to identify questions that could guide research beyond that domain.

39 Future Work

Several directions follow naturally.

39.1 Microscopic Model Construction

Define explicit microscopic variables

$$\chi_i$$

and their dynamics.

39.2 Emergent Metric Reconstruction

Construct an algorithm

$$\mathcal{C}[\chi] \rightarrow g_{\mu\nu}.$$

39.3 Causal Emergence

Investigate whether causal ordering can produce effective Lorentzian geometry.

39.4 Effective Action Derivation

Derive

$$S_{\text{EH}}$$

from microscopic dynamics rather than assuming it.

39.5 Quantum-to-Classical Transition

Determine how smooth spacetime emerges from microscopic quantum degrees of freedom.

39.6 Experimental Signatures

Search for measurable deviations from general relativity or quantum field theory in regimes where the microscopic theory predicts them.

40 Conclusion

This paper has developed a serious methodological question:

How much of our description of reality is empirically forced, and how much might be an effective representation of something deeper?

Einstein's theories demonstrate the power of principled assumptions. Special relativity establishes a profound connection between space, time, and Lorentz symmetry. General relativity establishes a profound connection between geometry and gravitation. The empirical success of these theories is not diminished by asking whether they are fundamental. Instead, their success provides the boundary conditions that any deeper theory must respect. The proposed framework of Assumption Stress Testing separates

assumptions $\rightarrow$ mathematics $\rightarrow$ predictions $\rightarrow$ evidence.

This separation makes it possible to question foundations without confusing speculation with established fact.

The speculative Information-Geometric Emergence hypothesis explored here proposes the possibility that

$$\mathcal{I} \rightarrow \mathcal{R} \rightarrow g_{\mu\nu}$$

could represent a deeper hierarchy beneath classical spacetime.

Nothing in the present work establishes that this hierarchy is physically correct.

Its purpose is to formulate the question precisely enough that a future theory could either support or reject it.

A genuine successor to relativity must satisfy a demanding condition:

New physics $\longrightarrow$ old physics in the validated limit.

It must reproduce special relativity.

It must reproduce general relativity.

It must preserve experimentally established causal structure.

It must explain the emergence of classical spacetime if spacetime is indeed emergent.

And, ultimately, it must make predictions.

The central methodological conclusion can therefore be expressed compactly:

Successful $\neq$ Fundamental

and

Incomplete $\neq$ Wrong.

Einstein's theories should not be treated as obstacles to deeper physics.
They should be treated as constraints imposed by nature.
If a deeper theory exists, it will not defeat relativity by simply contradicting it.
It will explain why relativity works.
That is the real “batsh*t insanity” worth pursuing:

Do not destroy the successful theory.

Find the structure underneath it.

References

- [1] A. Einstein, “On the Electrodynamics of Moving Bodies,” *Annalen der Physik*, Vol. 17, pp. 891–921, 1905.
- [2] A. Einstein, “Does the Inertia of a Body Depend Upon Its Energy-Content?” *Annalen der Physik*, Vol. 18, pp. 639–641, 1905.
- [3] A. Einstein, “On the Relativity Principle and the Conclusions Drawn from It,” *Jahrbuch der Radioaktivität und Elektronik*, Vol. 4, pp. 411–462, 1907.
- [4] A. Einstein, “The Field Equations of Gravitation,” *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, pp. 844–847, 1915.
- [5] A. Einstein, “The Foundation of the General Theory of Relativity,” *Annalen der Physik*, Vol. 49, pp. 769–822, 1916.
- [6] A. Einstein, *Relativity: The Special and General Theory*, Methuen, London, 1920.
- [7] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*, W. H. Freeman, San Francisco, 1973.
- [8] R. M. Wald, *General Relativity*, University of Chicago Press, Chicago, 1984.
- [9] S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, Addison-Wesley, San Francisco, 2004.
- [10] K. S. Thorne, *Black Holes and Time Warps: Einstein's Outrageous Legacy*, W. W. Norton, New York, 2017.